KOÇ ÜNİVERSİTESİ

MECH 203 Project Report

GROUP 3: Gasoline Generator



Group members

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TA: Selin Sude Kavun

Introduction

1. What is your project?

Our project is a **Gasoline Generator** designed to convert mechanical energy into electrical energy using gasoline as fuel. Internal combustion in its engine is the central part of creating motion to drive an alternator, which produces electricity through electromagnetic induction.

2. What is it used for?

The gasoline generator is a portable electrical energy source, particularly useful in areas without access to the primary power grid or during power outages. It finds applications in homes, outdoor events, construction sites, and emergency services.

3. How does it operate?

The Gasoline Generator can be divided into five important sub-parts:

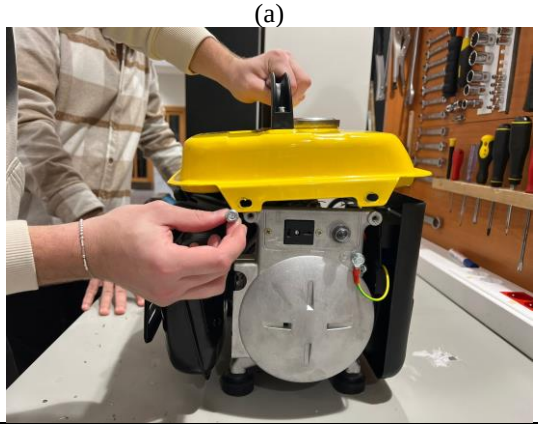
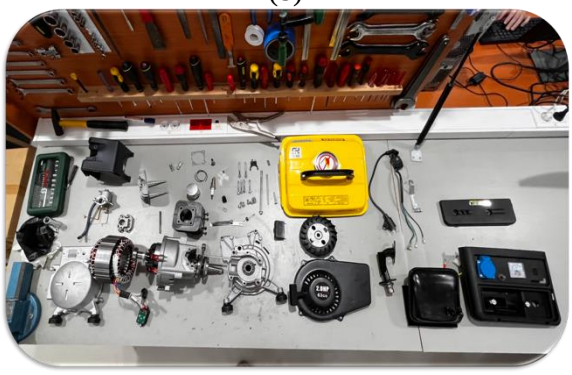
1. **Engine:** Converts the chemical energy of gasoline into mechanical energy through internal combustion.
2. **Alternator:** Transforms mechanical energy into electrical energy via electromagnetic induction.
3. **Fuel System:** Supplies gasoline to the engine.
4. **Cooling & Exhaust Systems:** Manage engine heat and emissions to prevent overheating, ensuring efficient and safe operation.
5. **Control Panel:** Provides a user interface for managing the generator's functions.

Photo of your object in real life	Picture of your complete and rendered assembly
	

Real photograph of the fully assembled project, and software-rendered image of the complete assembled project.

4. *Disassembly (All Groups Members)*

The primary challenge encountered during the disassembly process was the removal of the engine parts. The tools available in the laboratory were not sufficient for this task, so the parts were taken to a workshop in Sarıyer, where they were successfully disassembled. During this process, tools such as calipers, radius gauges, protractors, and compasses were used, and practical solutions were developed where these tools proved insufficient. Also, when disassembling the generator, we made sure to record a video so that we know how they looked before we took them apart.

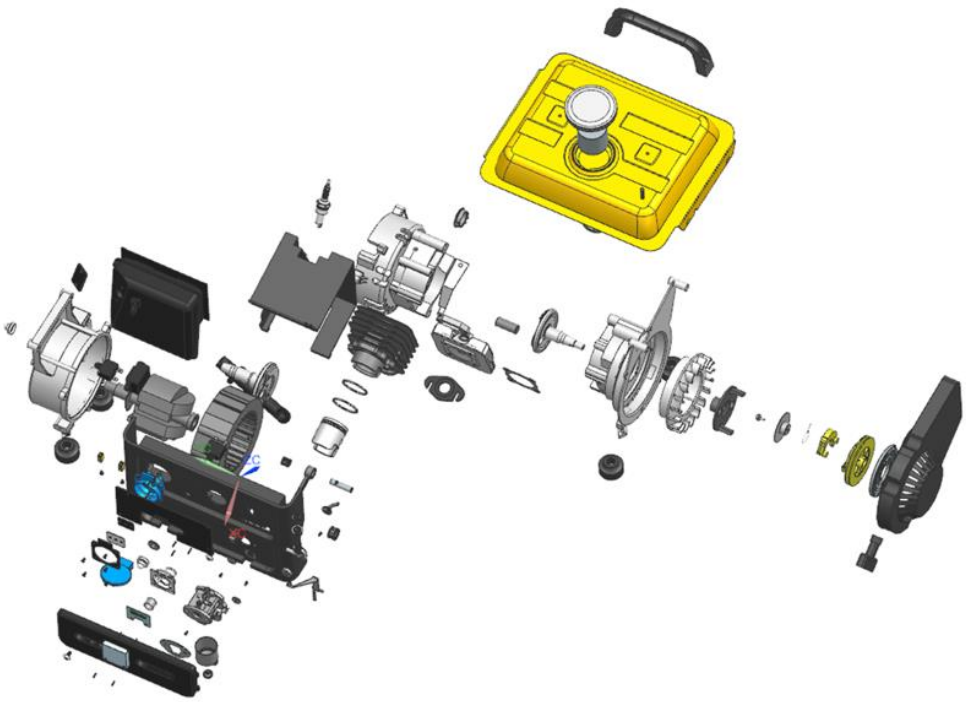
Disassembly photo	Disassembly photo
(a) 	(b) 
Fig. 2. (a)Disassembly process of the product by group members (b) disassembled parts	

5. *Assembly and Exploded View (Furkan Özgümüş)*

The generator consists of 111 different parts, which were all assembled during the assembly process. Almost all of the constraints, such as align, fix, concentric, parallel, and distance, were used. These constraints were necessary due to the large number of parts in the generator and their varying shapes. The biggest challenge faced during the assembly was that it was not checked whether the parts were compatible with those of others during their design. As a result, many parts did not fit together properly and had to be redrawn. For this reason, the assembly process was completed in 20 hours.



Complete assembled product in Siemens NX



Exploded View

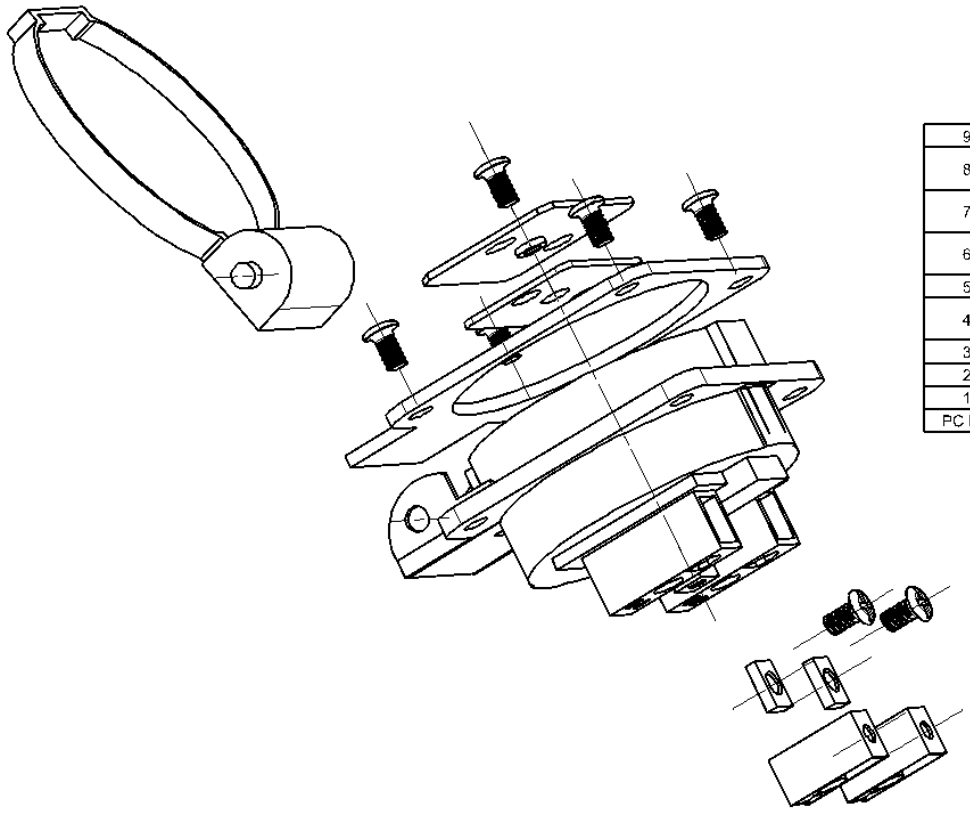
6. PLUG (Designer: Abdulsamet Bolat)

The plug is a vital part of the generator's electrical system. It provides the secure connection of electrical devices to the generator. During the design of Siemens NX:

- **Commands Used:** Tools such as "Extrude," "Holes," "Edge Blends," and "Shell" were used to redesign the plug.
- **Dimensions:** The drafting for this part is important to ensure everything fits and functions correctly. A perpendicularity tolerance on surface **C** ensures the plug aligns properly with the socket, preventing any misfits or poor connections. Parallelism on surfaces **A** and **B** keeps them aligned and even, ensuring the plug fits securely without being too loose or too tight. These tolerances were selected to control the critical dimensions needed for reliable performance and proper assembly with the socket.

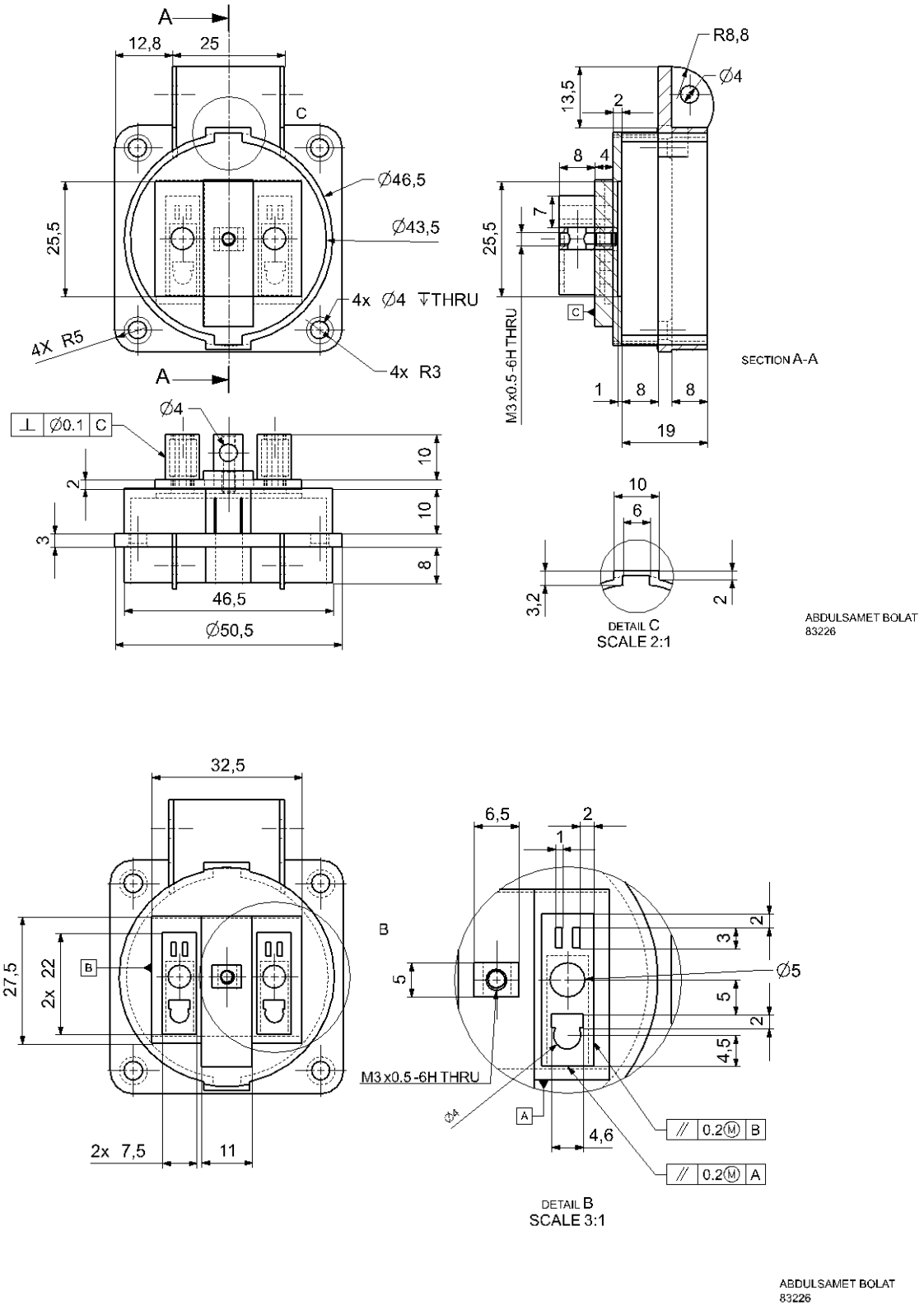
Designed part in NX	Real part
(a) 	(b) 
Comparison between (a)Designed part in Siemens NX and (b) real part photo	

Drafting of the designed part



PC NO	PART NAME	QTY
9	PLUG_COVER	1
8	PLUG_SPACER_WHITE	1
7	PLUG_SPACER_BLACK	1
6	PLUG_SUBASSEMBLY_CONNECTOR	2
5	PLUG_CONNECTOR	2
4	PLUG_PART_FOR_CONNECTOR	2
3	PLUG_SCREW	7
2	PLUG_SEAL	1
1	PLUG_BODY	1

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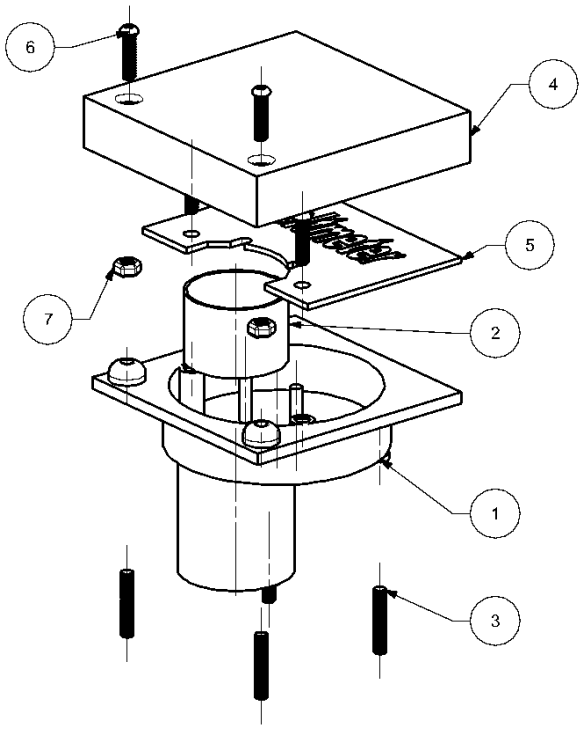
Detailed drafting of designed *plug*

7. Voltmeter (Designer: Abdulsamet Bolat)

The voltmeter is a critical component located on the control panel of the gasoline generator. It enables users to monitor the voltage output of the system, ensuring safe and efficient operation. As one of the key elements of the control interface, the voltmeter provides real-time feedback that helps prevent overvoltage or undervoltage conditions, maintaining the stability and reliability of the generator. Its design prioritizes precision and proper integration with both each subpart and the plastic main body. Using Siemens NX, the voltmeter and its subcomponents were modeled with a focus on detailed tolerancing and alignment to ensure seamless assembly and functionality.

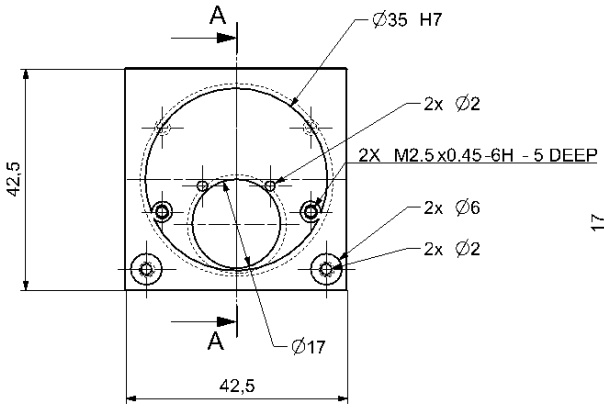
Designed part in NX	Real part
(a)	(b)
	
Comparison between (a)Designed part in Siemens NX and (b) real part photo	

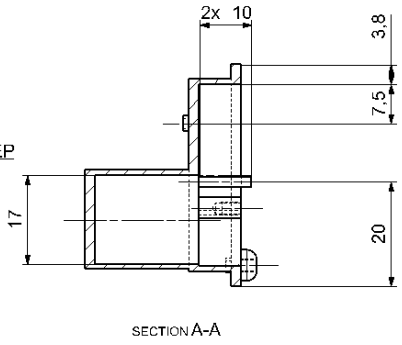
Drafting of the designed part



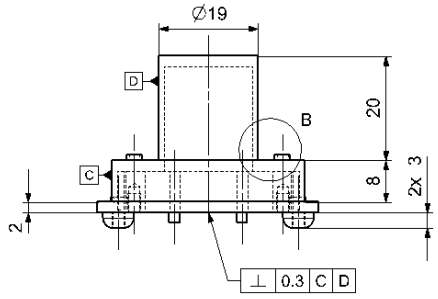
7	VOLTMETER_NUT	2
6	VOLTMETER_SCREW	4
5	VOLTMETER_DISPLAY	1
4	VOLTMETER_COVER	1
3	VOLTMETER_NUT_SCREW	4
2	VOLTMETER_CYLINDER	1
1	VOLTMETER_BODY	1
PC NO	PART NAME	QTY

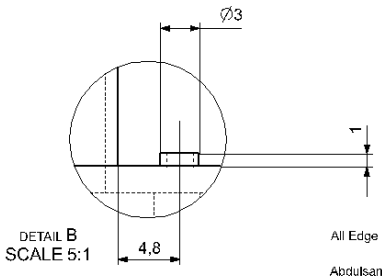
Abdulsamet Bolat
83226





SECTION A-A





DETAIL B
SCALE 5:1

All Edge Blends are R2

Abdulsamet Bolat
83226

Detailed drafting of designed *VOLTMETER*

8. *Fuel Tank (Designer: Yiğit Berk Uçar)*

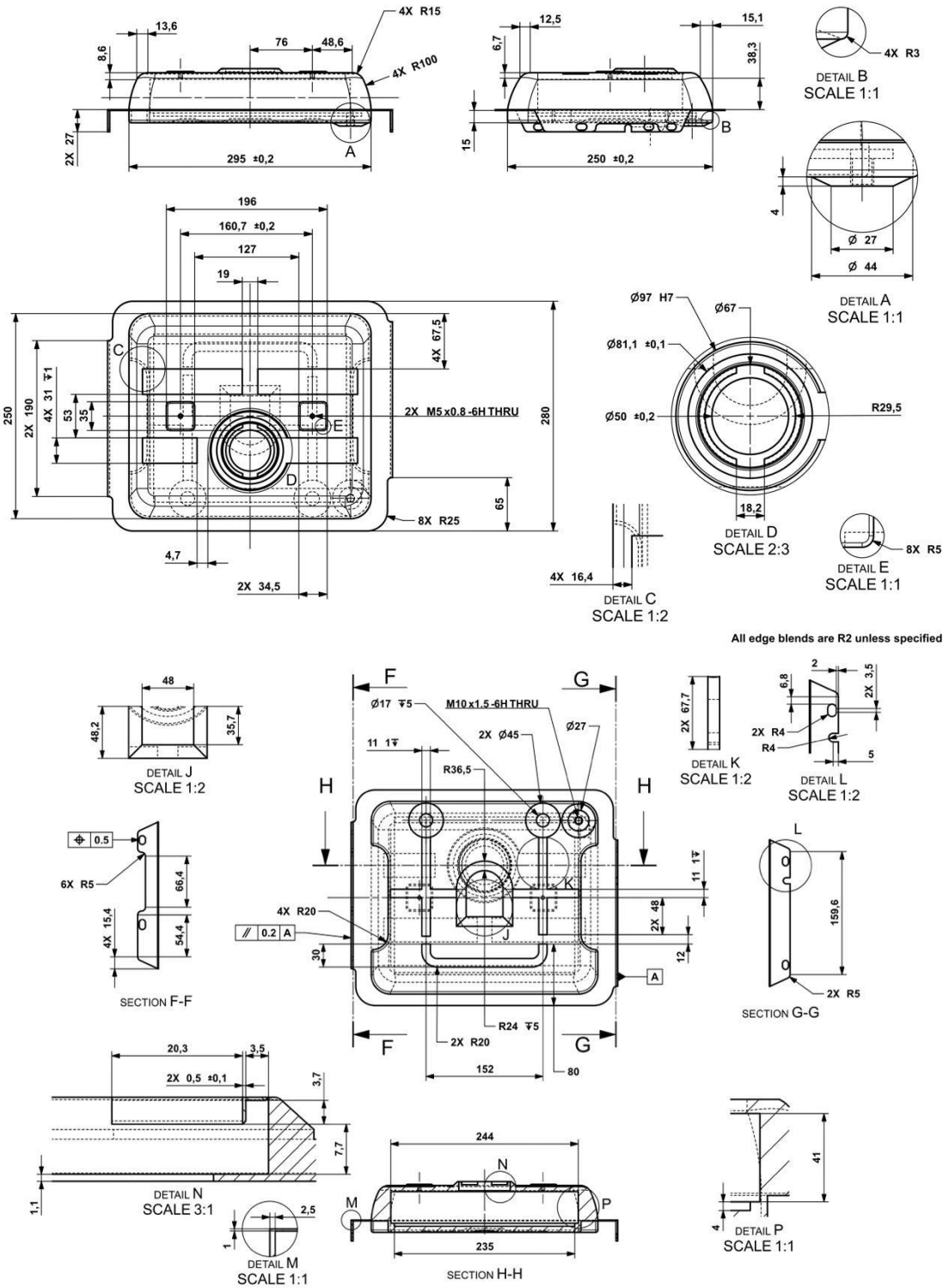
Drafting for the fuel tank was essential to ensure the design could be accurately manufactured and assembled. It provided detailed information about the critical dimensions, tolerances, and constraints, which are vital for leak-proof performance and proper alignment with other components, such as the electric motor body.

Designed part in NX	Real part
	

Fig. 5. Comparison between (a)Designed part in Siemens NX and (b) real part photo

Surface and dimensional tolerances were carefully chosen for optimal functionality. The fuel filler neck was assigned an H7 tolerance and the fuel cap an H6 tolerance to achieve a secure, leak-proof transition fit. Additionally, position and parallelism tolerances were applied to maintain proper alignment with the motor body, ensuring seamless assembly and reliable operation.

Drafting of the designed part (Sheet 1 - Sheet 2)



Detailed drafting of designed *Fuel Tank*

9. Alternator Cover (Designer: Yiğit Berk Uçar)

Drafting for the alternator cover was crucial as it ensured that the design could be manufactured accurately while meeting functional requirements. The detailed sketches accounted for the asymmetrical shape and the need for precise alignment with surrounding components, such as the stator and cover parts. This was vital for maintaining protection, reducing vibration, and ensuring rigidity.

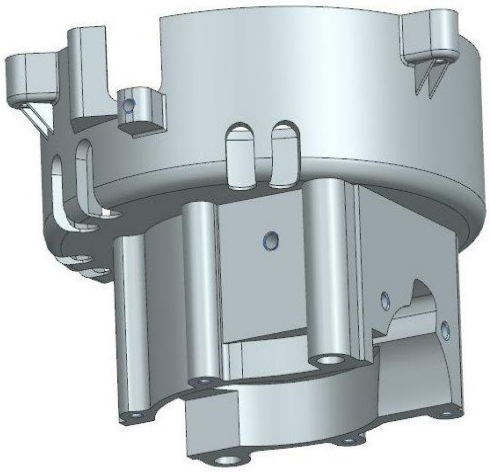
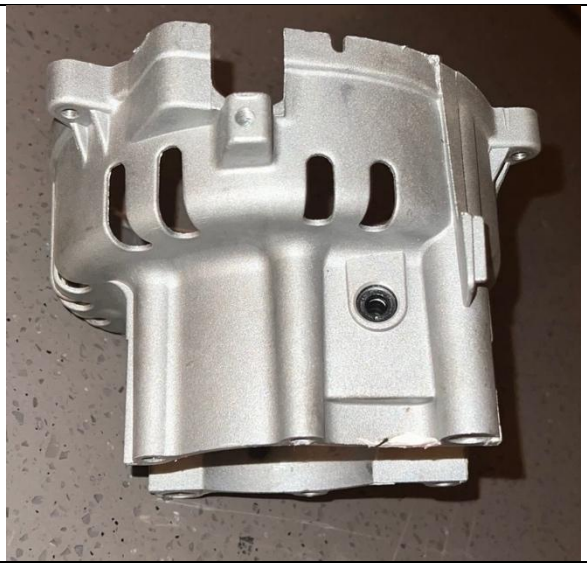
Designed part in NX	Real part
	

Fig. 5. Comparison between (a)Designed part in Siemens NX and (b) real part photo

Surface and dimensional tolerances were carefully applied to ensure proper alignment and fit. Parallelism tolerance was assigned to ensure alignment between the stator and cover parts, enabling precise fitting of the carburetor and piston components. Additionally, concentricity tolerance was used to align the hole's center with the crankshaft center, ensuring balance and rotational accuracy.

Drafting of the designed part (Sheet 1 – Sheet 2)

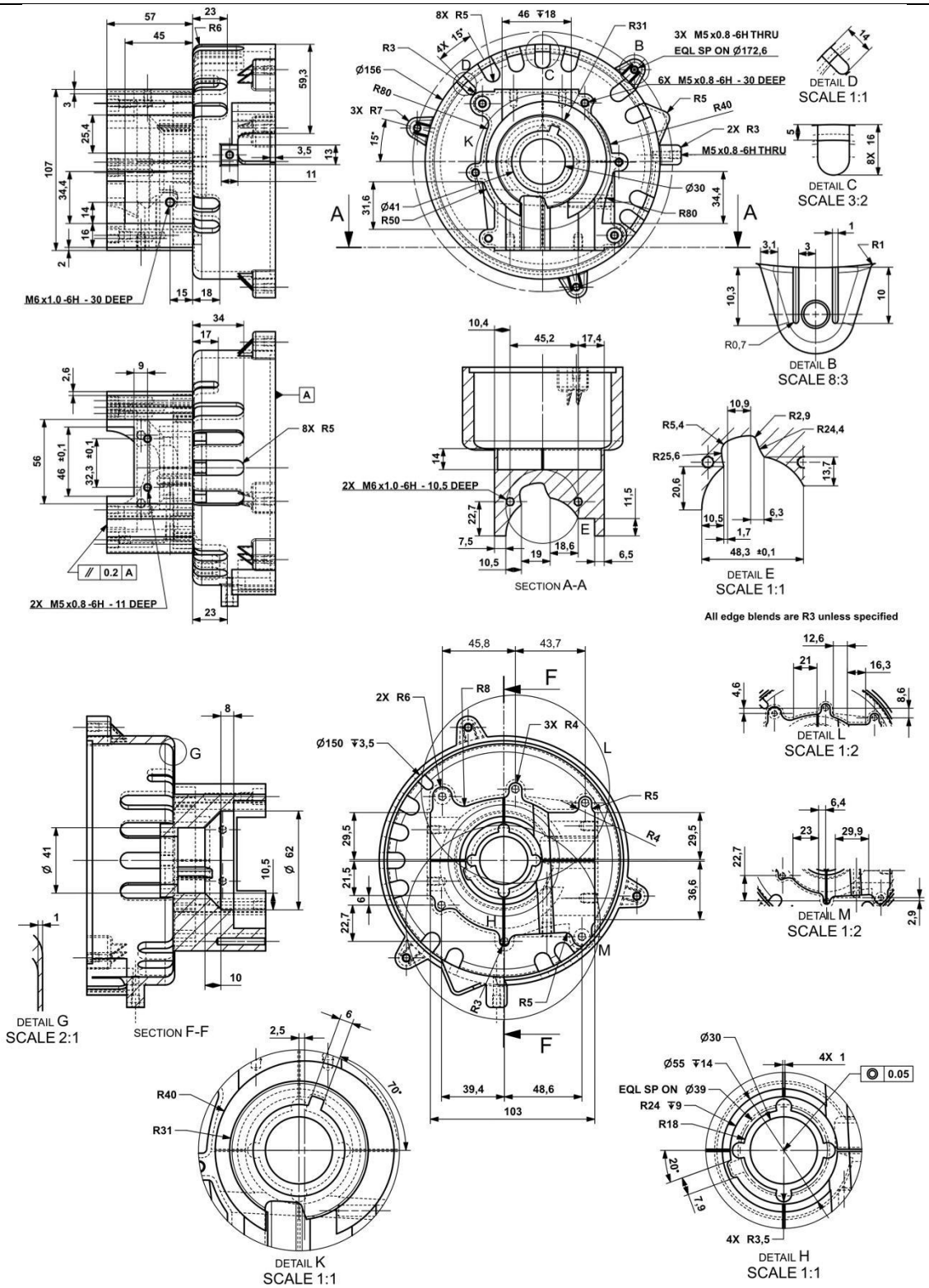


Fig. 6. Detailed drafting of designed *Alternator Cover*

10. Electric Motor Body Cover (Designer: Orkun Zengin)

The electric motor cover not only protects the motor from external factors but also facilitates the release of hot air from the system through its air vents. Additionally, its mounting feet provide structural support to the system. During the design process of the part, *revolve*, *extrude*, and *thread* commands were utilized. Due to its asymmetrical structure and small-scale details, the part has a highly detailed draft. This part includes parallelism and thread tolerances.

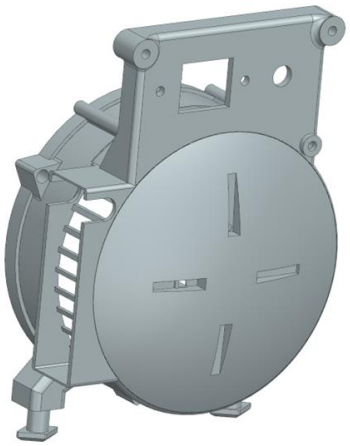
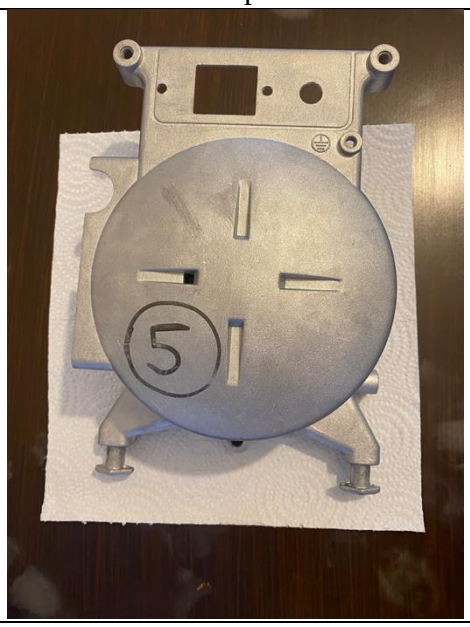
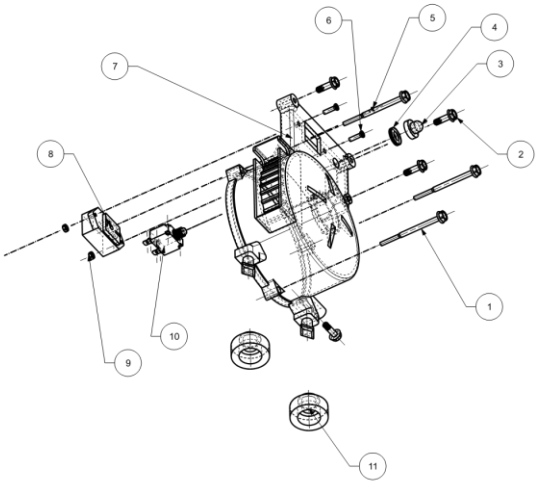
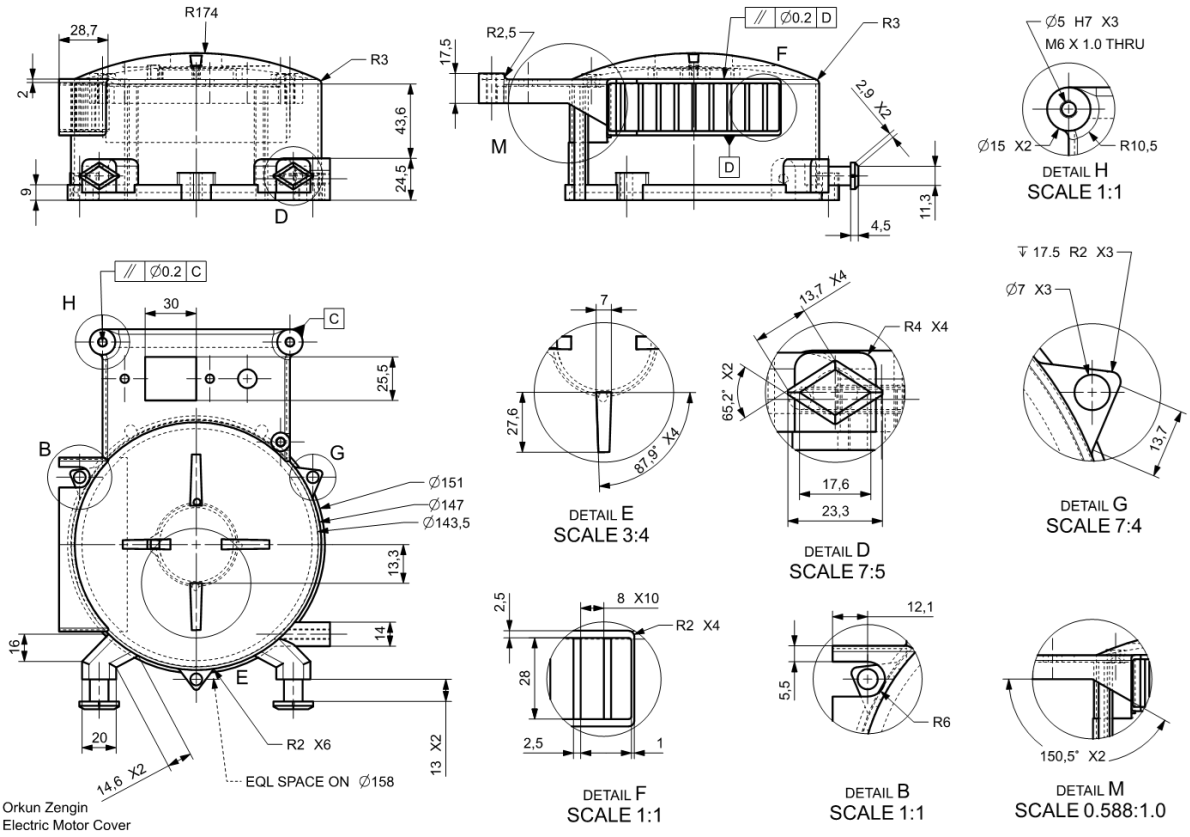
Designed part in NX	Real part
	

Fig. 5. Comparison between (a)Designed part in Siemens NX and (b) real part photo

Drafting of the designed part



PC NO	PART NAME	QTY
1	BOLT 10MM 81MM X3	3
2	BOLT 21.3 7.4MM M6	4
3	TRANSPARENT COVER OF BUTTON	1
4	BUTTON PLASTIC NUT	1
5	BOLT 6MM 5 MM	1
6	BOLT (B 7MM 16MM M6)	2
7	ELECTRIC MOTOR BODY	1
8	ELECTRONIC PART OF THE BODY	1
9	LOAF (B 7 MM 16MM M6)	2
10	BUTTON COMPONENT	1
11	ELECTRIC MOTOR BODY STAYS	2



Orkun Zengin
Electric Motor Cover

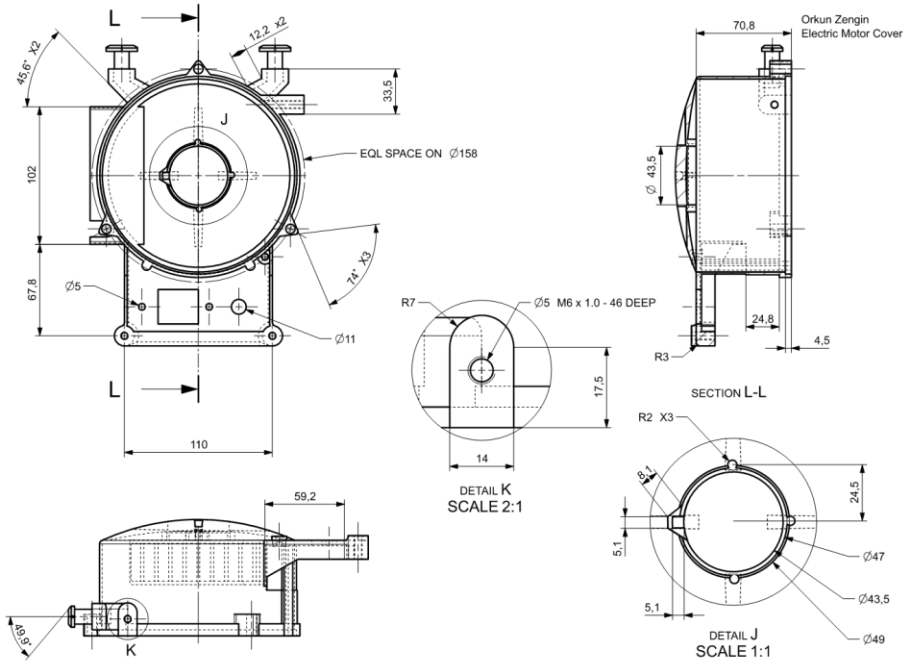


Fig. 6. Detailed drafting of designed *Electric Motor Body Cover*

11. Air Inlet (Designer: Orkun Zengin)

The air intake, with its grid-like openings, is responsible for drawing air into the entire system, thereby cooling the whole structure. It is one of the main components of the operating mechanism and is located at the center of the sub-assembly. The modeling of this part frequently utilized commands such as *revolve*, *extrude*, and *edge blend*. A positional tolerance was applied to the section directly connected to the shaft, while parallelism and angularity tolerances were assigned to the parts interfacing with other components.

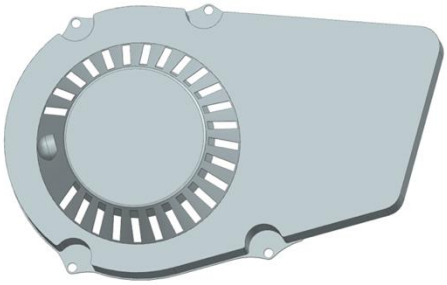
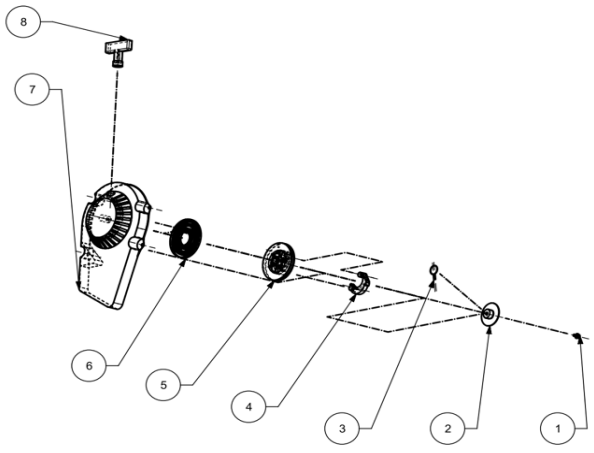
Designed part in NX	Real part
	

Fig. 5. Comparison between (a)Designed part in Siemens NX and (b) real part photo

Drafting of the designed part


PC NO	PART NAME	QTY
1	Bolt 8mm 13mm	1
2	Metal Part of The Starting Mechanism	1
3	Circle of The Metal Part	1
4	Rotating Part of The Disc	1
5	Disc	1
6	Band Spring	1
7	Air Inlet	1
8	Starting Arm	1

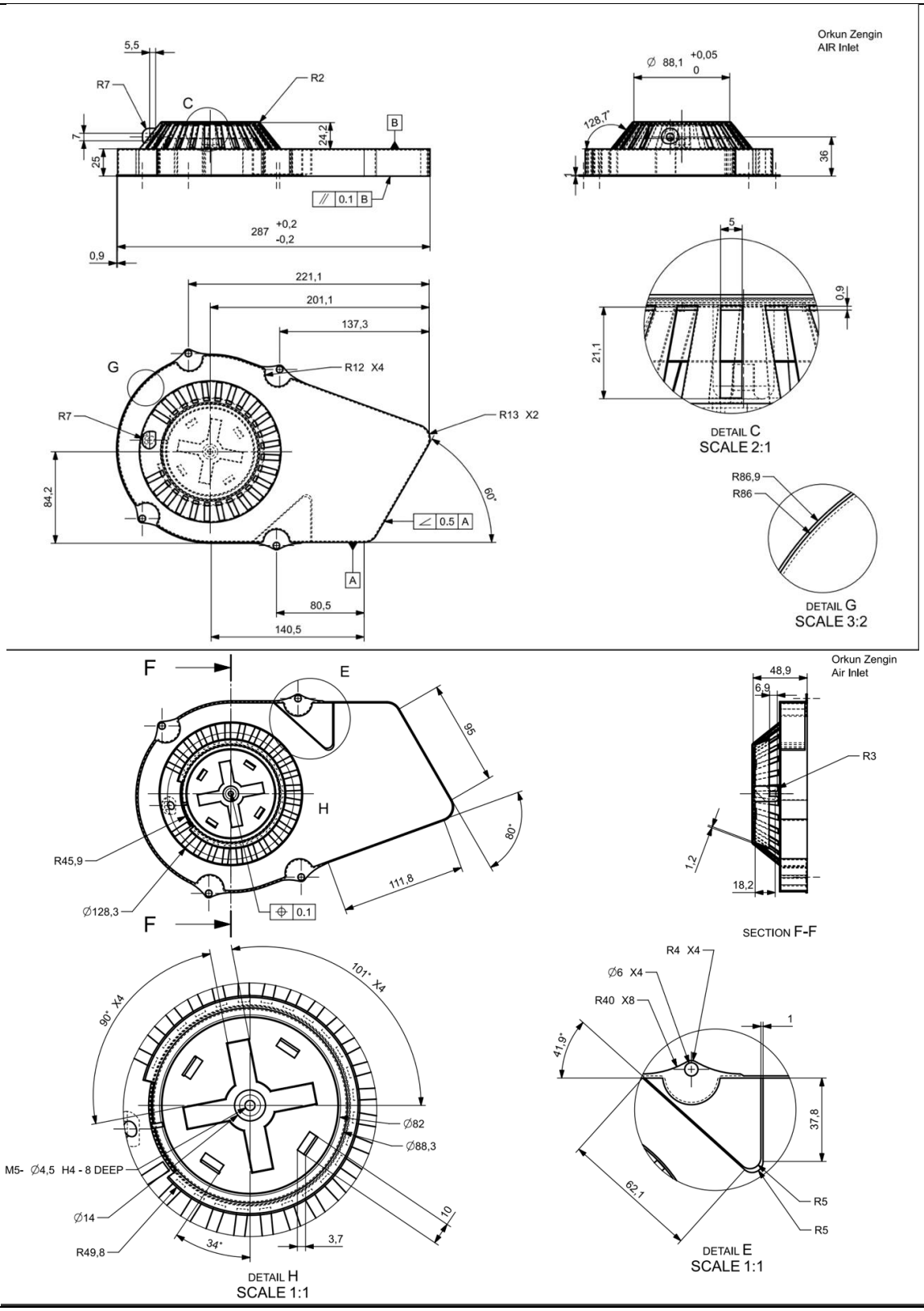
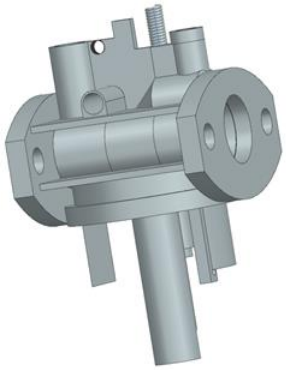
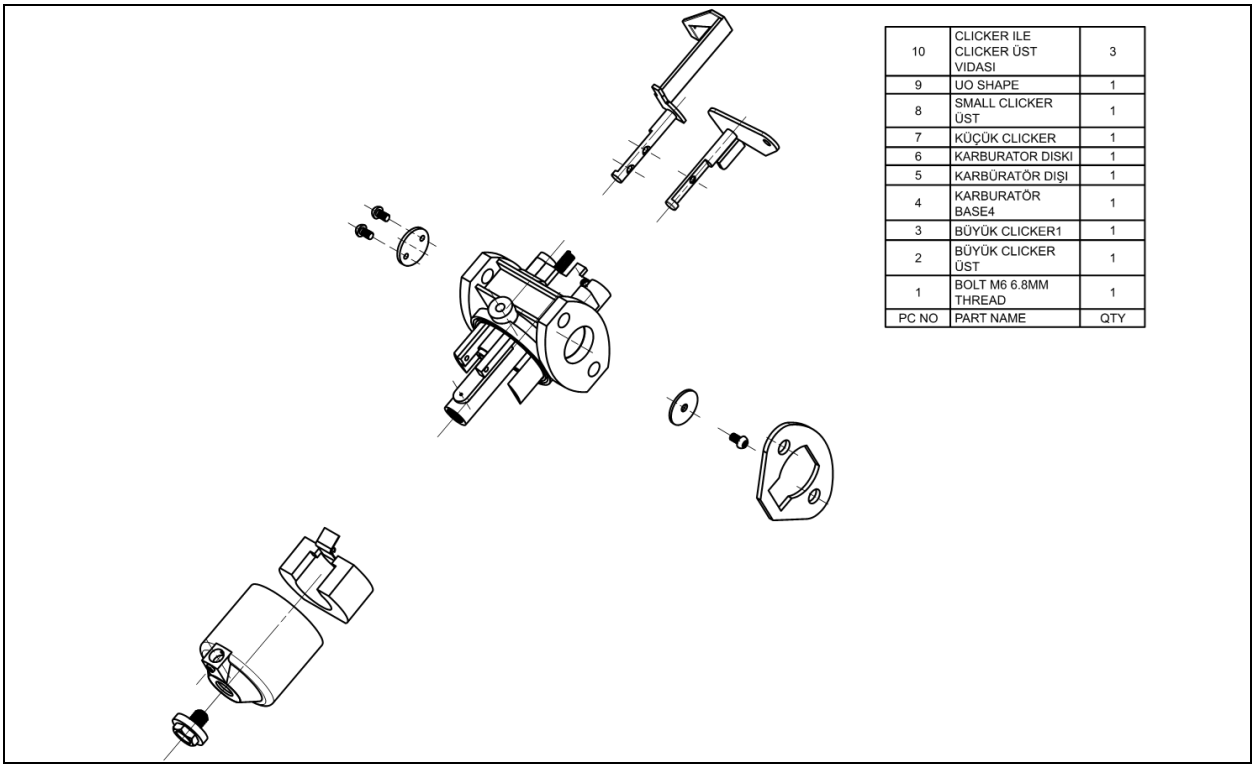


Fig. 6. Detailed drafting of designed *Air Inlet*

12. Carburetor (Designer: Bertan Erman)

The primary function of a carburetor is to mix air and fuel in the correct ratio to ensure efficient combustion in the engine. The main hole narrows in the middle, creating a venturi effect to regulate airflow and fuel delivery. This part of the carburetor is manufactured using CNC machines to achieve precise measurements. The design process began with parallel planes, followed by creating circles and adding intricate details. Additionally, a helix structure was incorporated into the design using helix and sweep commands. (In the files that are uploaded to the LearnHub, this part is named as 'karburator base drafting4', and it cannot be renamed since when it is changed, the assembly will not open).

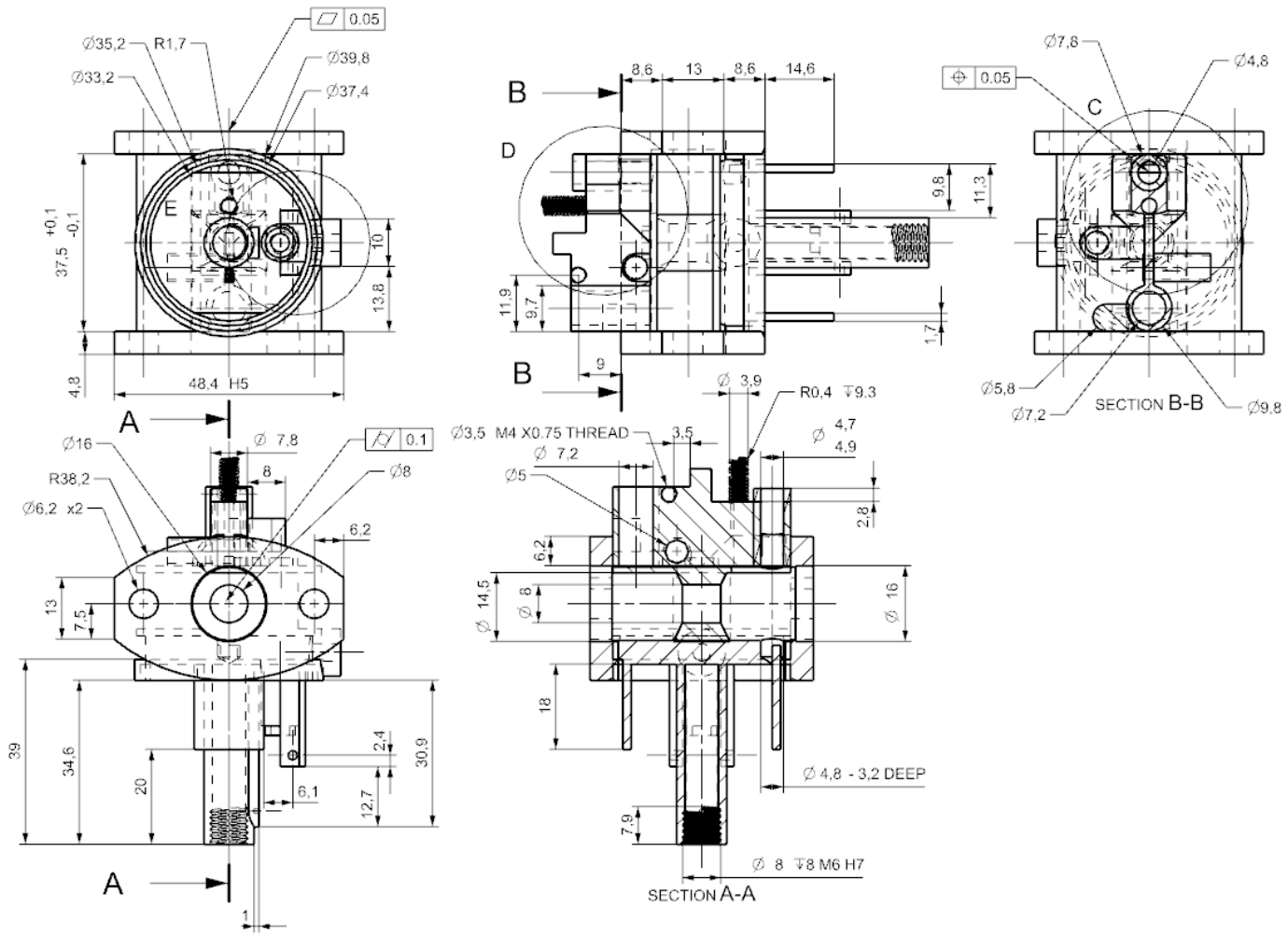
Designed part in NX	Real part
	
(a)	(b)
Comparison between (a)Designed part in Siemens NX and (b) real part photo	

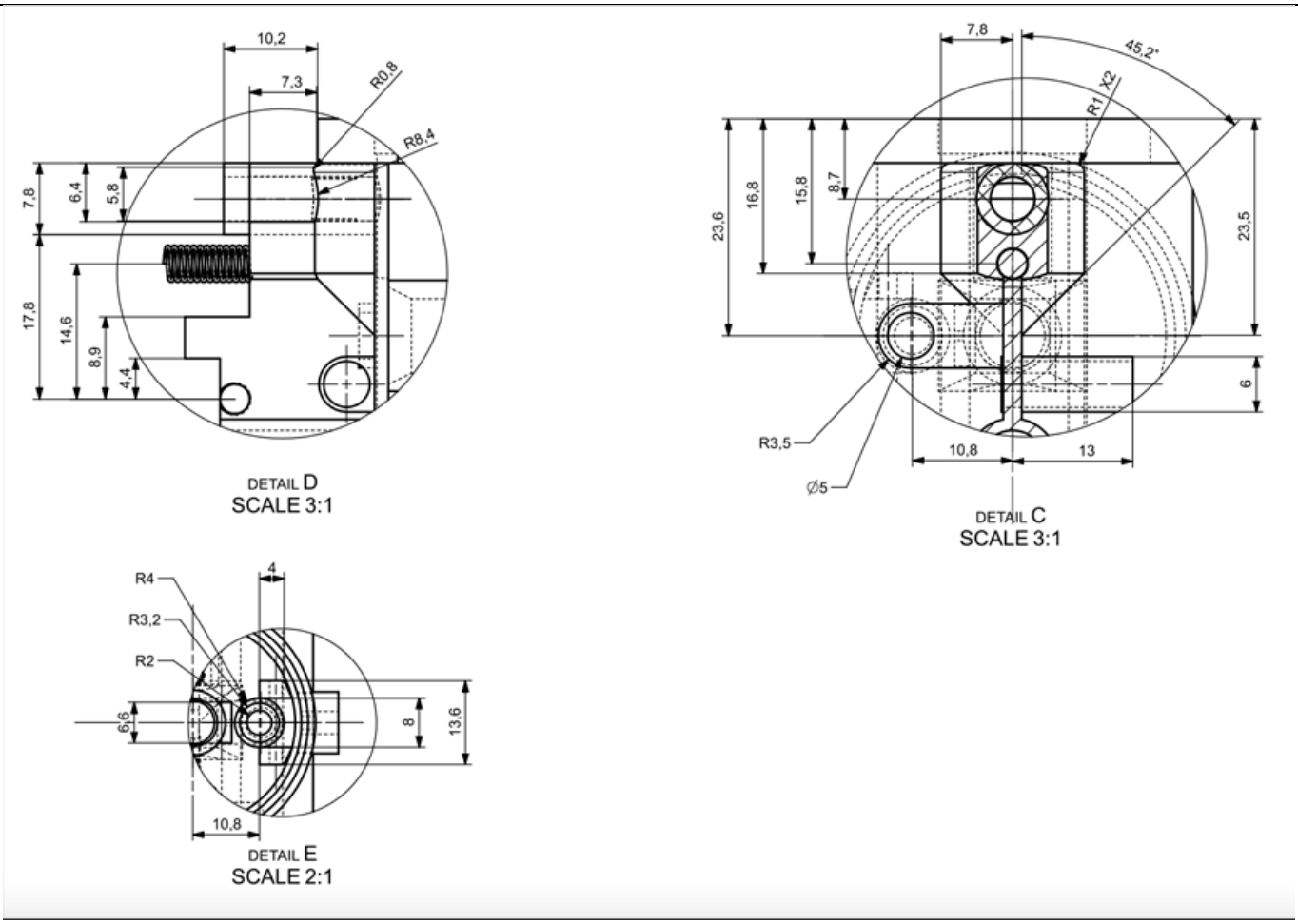


PC NO	PART NAME	QTY
10	CLICKER ILE CLICKER ÜST VIDASI	3
9	UO SHAPE	1
8	SMALL CLICKER ÜST	1
7	KÜÇÜK CLICKER	1
6	KARBURATOR DISKI	1
5	KARBÜRATÖR DIŞI	1
4	KARBURATÖR BASE4	1
3	BÜYÜK CLICKER1	1
2	BÜYÜK CLICKER ÜST	1
1	BOLT M6 6.8MM THREAD	1

In the middle of the part, there is a hole that needs to be precisely manufactured, so specific tolerances were applied. Cylindrical form tolerance is critical to ensure proper airflow as intended. Additionally, parallelism tolerance was specified (similar to the crank mount on the engine side) to maintain the required angle alignment.

Drafting of the designed part

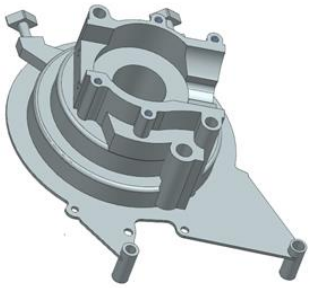




Detailed drafting of designed Carburetor

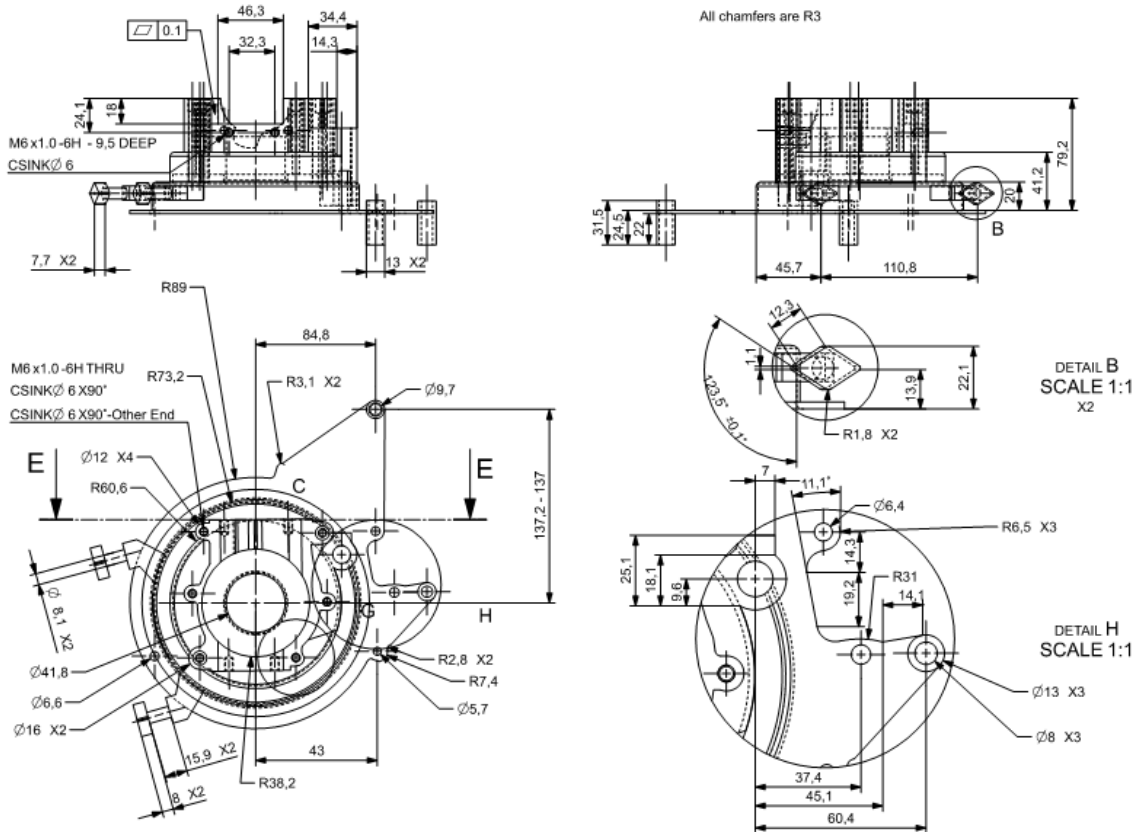
13. Crank mount engine side (Designer: Bertan Erman)

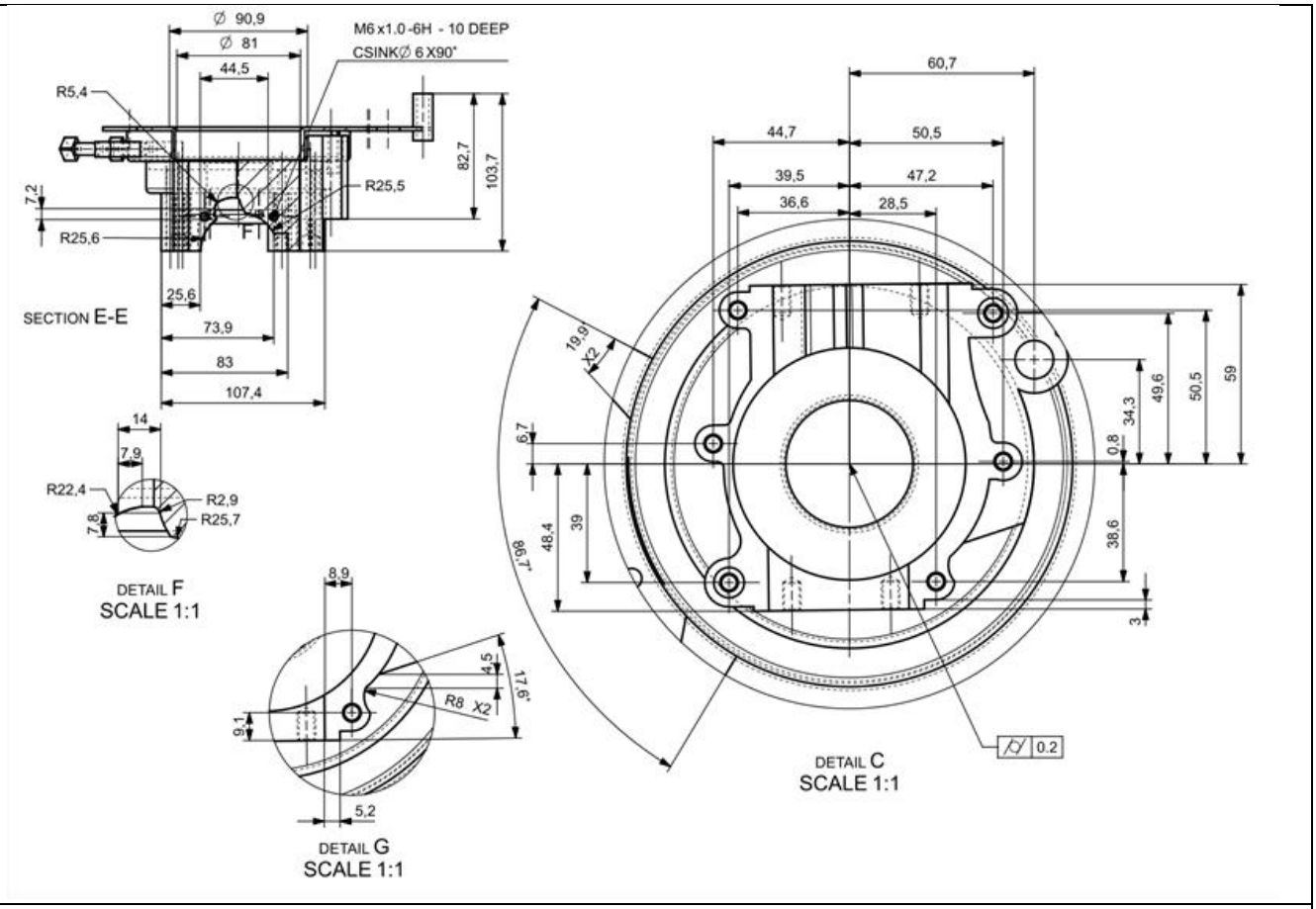
This part was asymmetrical and required three iterations of design to achieve precise modeling using commands such as extrude and revolve. Initially, the main challenge was determining the exact positions of the holes. To address this, a piece of paper was placed under the part, and a pen was used to mark the hole locations. This method allowed for accurate positioning, making it easier to draw the part. The holes at the edges were positioned based on input from other group members (In the files that are uploaded to the LearnHub, this part is named as 'Krank ana kapak yigide göre düzenlenmis hali(1)', and it cannot be renamed since when it is changed, the assembly will not open).

Designed part in NX	Real part
	
(a)	(b)
Comparison between (a)Designed part in Siemens NX and (b) real part photo	

After measuring the positions of the holes, the drafting process became straightforward. Hole tolerances were defined based on the requirements of the other components. Angle tolerances for the feet were crucial to ensure the generator remained level. Cylindrical form tolerances were applied to allow the shaft to pass through smoothly without causing wobbling effects.

Drafting of the designed part

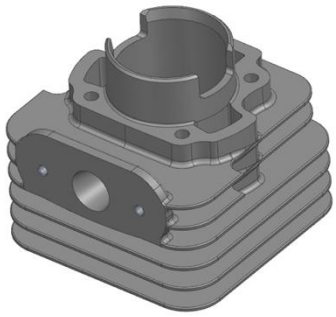




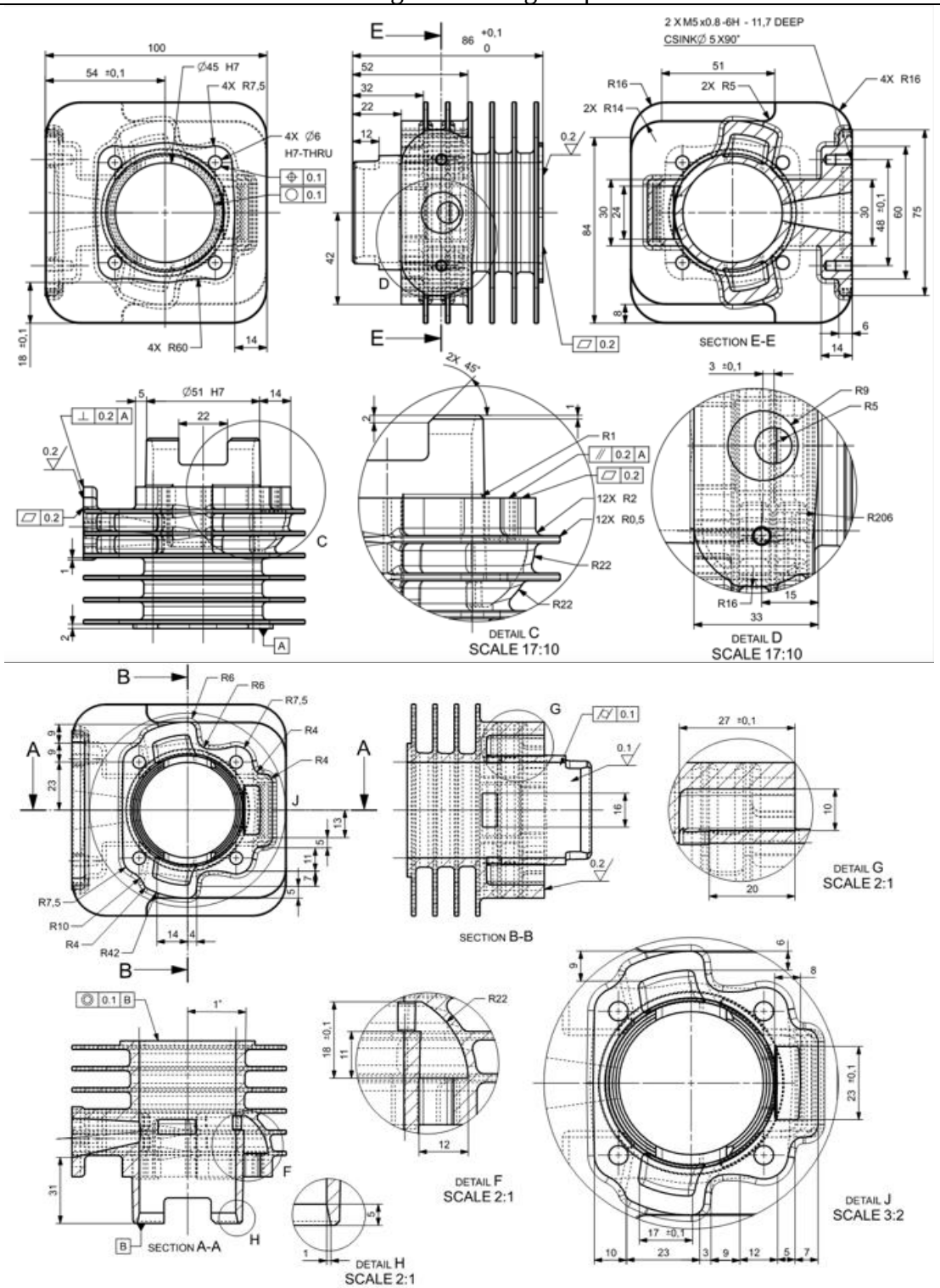
Detailed drafting of designed Crank mount engine side

14. Cylinder Block (Designer: Furkan Özgümüş)

One of the key parts in assembly is the cylinder block, which required precise modeling using commands like extrude, revolve, and fillet. To ensure proper alignment, constraints such as concentric and parallel were applied between critical features like the piston bore and mounting holes. Extra features like chamfers and ribs were added to improve durability and ease of assembly. The drafting process for this part was crucial to ensure accurate manufacturing and fit. Surface tolerances of 0.2 mm and dimensional tolerances of ±0.1 mm were assigned to critical surfaces like the piston bore and mounting interfaces to maintain functionality and compatibility with other parts.

Designed part in NX	Real part
	
Comparison between Designed part in Siemens NX and real part photo	

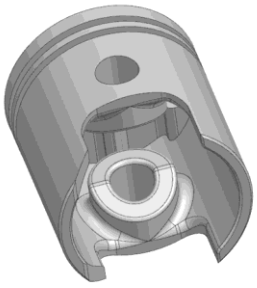
Drafting of the designed part



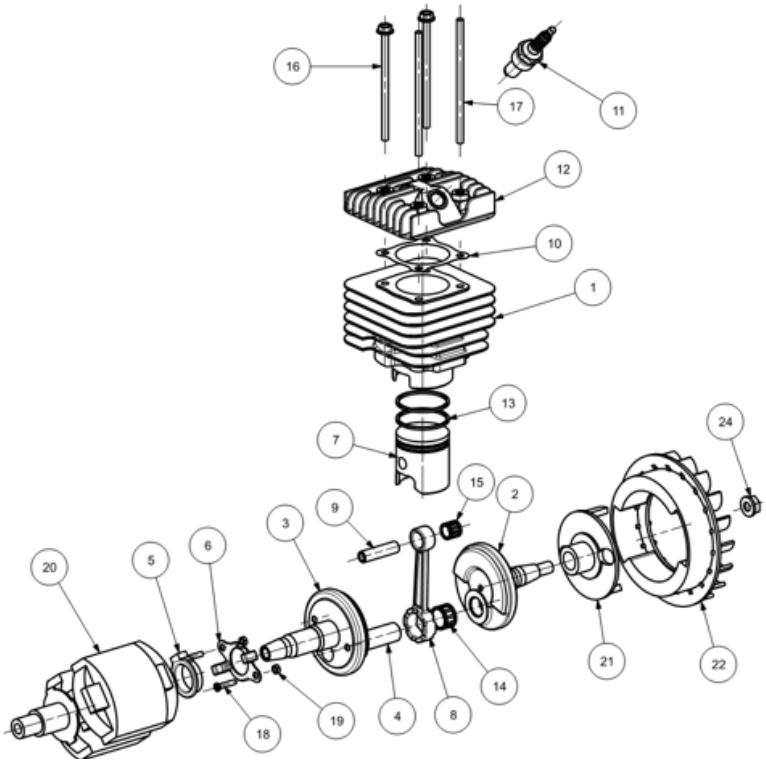
Detailed drafting of designed Cylinder Block sheet1/sheet2

15. Piston (Designer: Furkan Özgümüş)

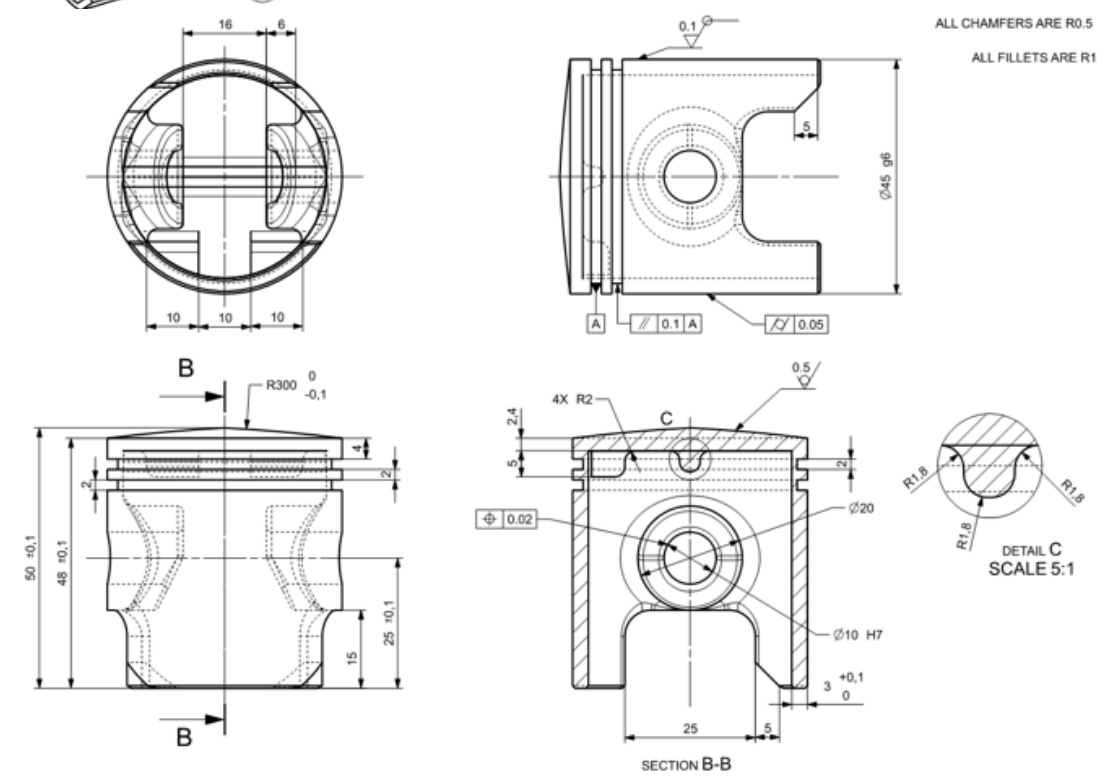
One of the critical parts in assembly is the piston, which was modeled using commands like extrude, revolve, and shell to achieve the hollow structure. Constraints such as concentric and coincident were applied to ensure alignment with the piston rod in the sub-assembly. Additional features like grooves for piston rings and fillets to reduce stress concentration were included for functionality and durability. The drafting process for this part is essential to ensure precise manufacturing, with surface tolerances of 0.1 mm on the piston walls for smooth operation and dimensional tolerances of ±0.05 mm on critical fits to guarantee compatibility with the cylinder.

Designed part in NX	Real part
	
Comparison between Designed part in Siemens NX and real part photo	

Drafting of the designed part



24	NUT_SHORT	1
23	ELEKTRIK MOTORU FANI	1
22	ELEKTRIKMOTORU(DIŞ KISIM BIÇAKLAR)	1
21	ELEKTRIKMOTORUF ANI(İÇ KISIM)	1
20	BOBIN IÇI 1H1	1
19	LOAF (B 7MM 16MM M6)1	2
18	BOLT 7MM 16MM M6 X2	2
17	THREAD_THREAD_1 13MM	2
16	NUT_THREAD_105M M	2
15	PISTON_BEARING	1
14	CRANK_BEARING	1
13	PISTON_RING	2
12	PISTON_HEAD	1
11	SPARK_PLUG	1
10	PISTON_GASKET	1
9	PISTON_BEARING_S HAFT	1
8	PISTON_ROD	1
7	PISTON	1
6	CRANK_PIECE2	1
5	CRANK_PIECE1	1
4	CRANK_BEARING_S HAFT	1
3	CRANK_ELECTRIC	1
2	CRANK_AIR	1
1	PISTON_WALL	1
PC NO	PART NAME	QTY

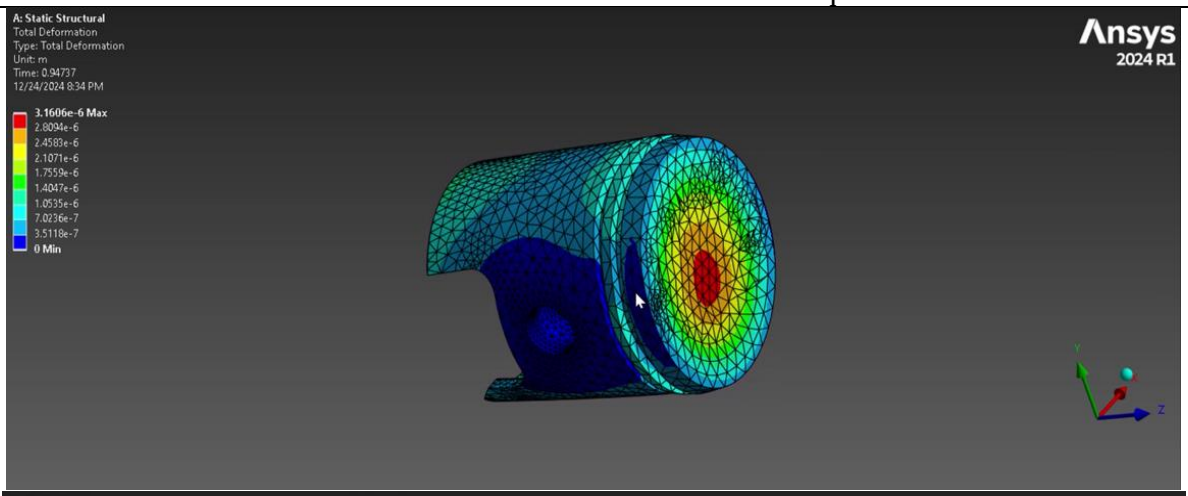


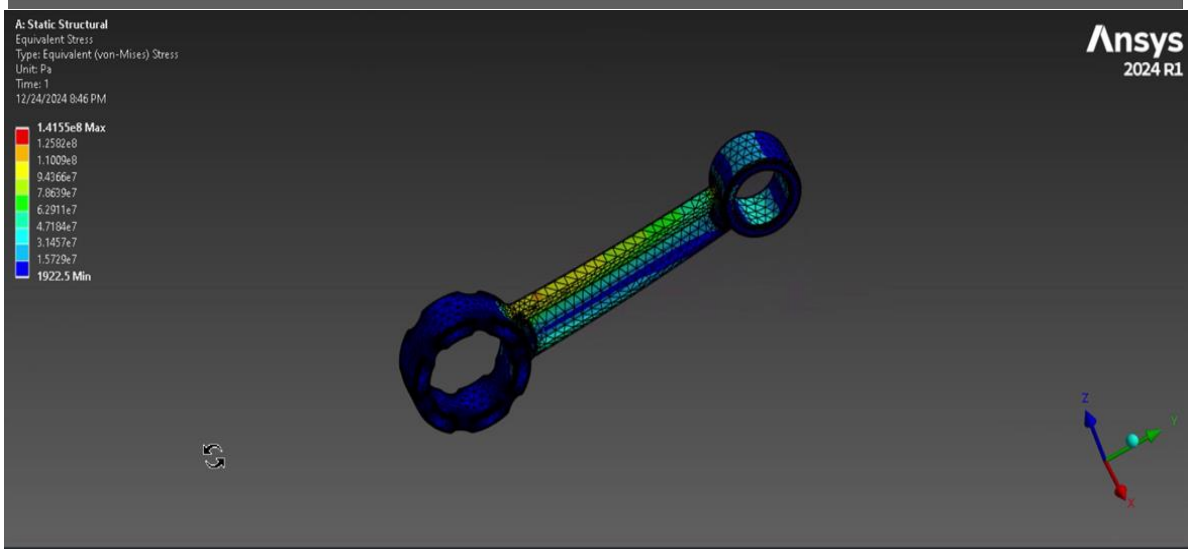
Detailed drafting of designed engine assembly exploded view and piston.

16. Finite Element Analysis (Furkan Özgümüş)

A finite element analysis was conducted on a piston and piston rod using static simulation. Aluminum was assigned as the material for the piston and steel for the rod, chosen for their mechanical properties and compatibility. Fixed constraints were applied at the rod's base, and a distributed load was applied on the piston head to mimic operating conditions. The results showed that stress was accumulated near the connection points, particularly at the rod's junction with the piston, indicating potential failure zones under high loading. This analysis was crucial for ensuring structural integrity and identifying areas for design reinforcement.

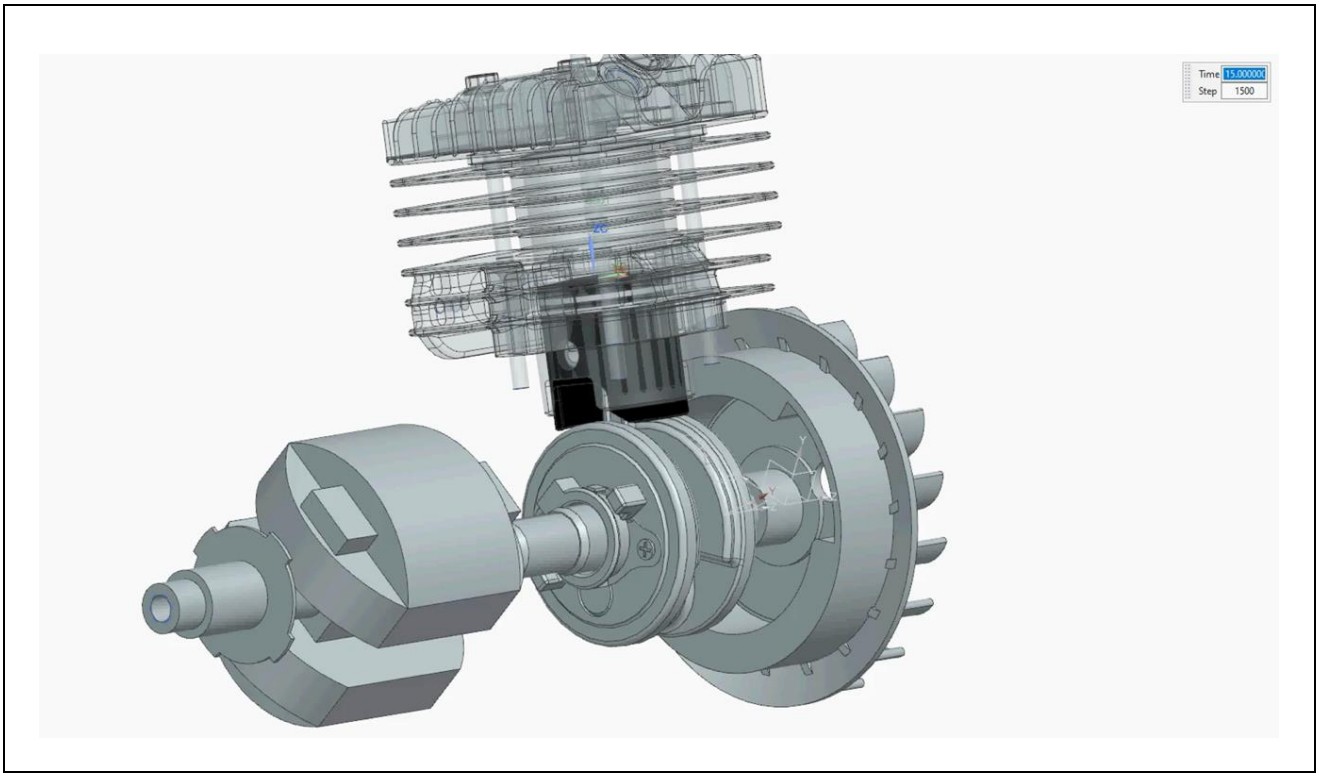
Piston: total deformation / Piston Rod: equivalent stress





17. Motion. (Bertan Erman, Yiğit Berk Uçar)

After the engine assembly was created, an attempt was made to simulate the motion using constraints; however, this approach was unsuccessful. A different approach was then tried, where the constraints were removed and replaced with motion bodies and fixes. Seven motion bodies and ten fixes were created, and an angular speed of 144 m/s was applied to the shaft. After the motion bodies and fixes were set up, several iterations were conducted to fine-tune the system and address any issues in the motion simulation. Adjustments were made to the angular velocity, and the connections between the components were verified to ensure smooth and synchronized operation. The expected behavior was demonstrated through the simulation results, and all components were shown to interact as intended. Through this iterative approach, a fully functional motion system was achieved, meeting the design and performance requirements effectively.



18. Presentation

This project presentation, fully prepared and completed by Abdulsamet Bolat.

19. Conclusion

Through this project, we gained valuable insights into the design, assembly, and functionality of a gasoline generator. We developed practical skills in 3D modeling, drafting, and tolerancing using Siemens NX. Also, We have faced problems during whole project and work as a team to solve them. The hands-on experience enhanced our understanding of mechanical systems and their integration. Additionally, conducting finite element analysis deepened our knowledge of structural integrity and stress management in mechanical parts. This project has significantly contributed to our engineering skill set and teamwork capabilities.